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Design of a graph-oriented database for data management in statistical data infrastructure

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ABSTRACT

In recent years, official statistical agencies have embraced the development of public statistics based on administrative records rather than surveys. This solution necessitates connecting numerous sources and silos to obtain not only the data but also their relationships, generating significant complexity in managing schemas and scalability. One of ISTAC's objectives (Canary Islands Institute of Statistics) is to promote an Integrated Data System (iDatos) to produce multi-source statistics. The large volume of data and its continuous growth requires the use of Big Data technology to ensure its efficiency and performance. The current solution stores information in a relational database with a design based on master data management. However, in recent years, the use of graph-based databases for master data management has increased in different contexts, where relationships constitute the central element of the data model. This paper presents the transformation of the current system to a design based on graph-based databases. This paper presents the analysis of the problem, the design of the graph-based database schema, and the implementation of a computational experiment to compare performance.

1. Introduction

In many countries in recent years, the production of official statistics based on surveys has been replaced by production through administrative records and even other Big Data sources. During the 21st century, this production method through data integration has been extended (Zhang, 2012). This practice was first consolidated in the Nordic countries, where most statistics were conducted using this type of system (Wallgren & Wallgren, 2015). This approach has been implemented over time by official statistical agencies with access to population registers and household censuses, facilitating the transition from a traditional production system to a register-based system (Solari et al., 2023). In the countries of the United Nations Economic Commission for Europe (UNECE), the traditional way of carrying out censuses has decreased between 2000 and 2020 (Department of Economic and Social Affairs, Statistics Division, 2022). For example, Germany has been following a register-based census model since 2011, supplemented by surveys used to correct errors or supplement information not otherwise available. The aim is to have a register-based census by 2031 (Söllner & Körner, 2022). In general, most European countries are moving toward the use of administrative registers for conducting censuses (EUSTAT , n.d.). In 2018, the Social Economy Commission for the Asia-Pacific region began work on adapting the UNECE data integration guide for official statistics to the Asia-Pacific community. In 2020, the Republic of Korea, Singapore, and Turkey conducted censuses based on register data, complemented by a large-scale survey. Canada has also made efforts to modernize its statistical infrastructure to link population, household, activity, and enterprise statistical registers. Population and household registers are used to support census development, and all of these registers can be utilized to produce other statistics. Trépanier (2022). With regard to Latin America, countries such as Mexico, Brazil, Colombia, Costa Rica, Chile, Ecuador, and Uruguay have developed or planned initiatives aimed at using administrative registers for statistical production (Wallgren & Wallgren, 2021). In Spain, the 2011 census combined administrative records with a survey (INE, 2011) and the 2021 census is based entirely on information from administrative records. The construction of the census involves filling in the information for census variables using data extracted from various administrative sources instead of conducting surveys with the population. Alternative sources have been utilised for some variables that cannot be obtained from administrative records. For example, in the 2011 census, an analysis of mobile phone location data was conducted to gather information about "daily mobility"; however, this variable was ultimately excluded (INE, 2019). On the other hand, the population count in the 2021 census is conducted using the "sign of life"

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method, determining from the population register whether an individual is a resident or not.

In the Autonomous Community of the Canary Islands, the ISTAC (Canary Islands Institute of Statistics) carries out the following statistical operations based on administrative records: the Spanish Labour Force Survey based on Registers (EPA-Reg, Encuesta de Población Activa con Registros) and the Labour Insertion Statistics (EIL, Estadística de Inserción Laboral). The objective of EPA-Reg is to obtain estimates of the population and their relationship with economic activity at the municipal and sub-municipal levels in the Canary Islands, based on place of residence (ISTAC, 2023; Yanes et al., 2018). This statistic provides data on economic activity related to the human component: it analyzes characteristics of different groups of people (active, employed, unemployed, and salaried by economic sector) and sociodemographic features on a quarterly basis. This is made possible through the georeferencing and integration of administrative data, such as the Municipal Register of Inhabitants (PMH, Padrones Municipales de Habitantes) (ISTAC, 2022), Social Security or General State Civil Servant Mutual Society (MUFACE, Mutualidad General de Funcionarios Civiles del Estado) affiliations, and records from the Canary Islands Employment Service, among others. EIL statistical variables are generated for quantitative studies on the relationship between qualifications offered in the Canary Islands (university and vocational training studies) and subsequent employment outcomes. In this context, enrollment and graduation records from Canary Islands universities are used, together with worker registration data in the Social Security and MUFACE systems, as well as records of graduates registered as employment seekers in the Canary Islands Employment Service and PMH.

Benefits of this model include reduced costs and burden on surveyed individuals, broader coverage, higher frequency, and more detailed statistics (United Nations Economic Commission for Europe, 2015), as well as its potential for conducting longitudinal studies and spatial analyses (Yanes, 2017). Moreover, once data are incorporated into the public statistics system, quality control and assurance ensure the reliability and integrity of official statistics, generating large volumes of valuable data that governments are beginning to exploit for intelligent data management and evidence-based decision-making. Recent advances in Artificial Intelligence engines-particularly in deep learning (LeCun et al., 2015), Transformers (Vaswani et al., 2017), generative AI (Rombach et al., 2022), deep reinforcement learning (Mnih et al., 2015), and explainability in machine learning algorithms (Doshi-Velez & Kim, 2017)-are contributing to the discovery of complex patterns in data across diverse domains of knowledge, such as AlphaFold (Guo et al., 2024) in molecular biology or imaging-based diagnostic systems in medicine (Guo et al., 2025). The adoption of these techniques by public sector organizations is driving the expansion of digital administrative data sources and is beginning to serve as a valuable tool to facilitate the production of high-quality information from vast and heterogeneous datasets (Valle-Cruz & García-Contreras, 2025). Ongoing efforts of public organizations to provide high-quality data will further facilitate research and the development of innovative AI-driven solutions of this kind, extending their potential applications to nearly every domain that affects the population. Among the disadvantages, we can mention dependency on the quality of administrative records, the maintenance of these records, and the fact that their management methodology is oriented toward administrative needs rather than statistical purposes. Finally, the use of individuals' personal information in statistics raises legal challenges.

ISTAC has focused its efforts on developing an administrative data bank for statistical purposes, aimed at facilitating file integration and the creation of multi-source statistics based on master data. Additionally, it has aimed to obtain spatial statistics through georeferencing within the Canary Spatial Statistics Framework.

This model of statistical production faces several technological challenges, including the adoption of new techniques such as record linkage and efforts to ensure data quality and traceability. It is necessary to address data inconsistencies, duplication, and inaccuracies from multiple sources for each statistical unit. This is not a new issue for organizations; in the last decade, Master Data Management (MDM) emerged as a strategy for managing shared data across different silos within the organization. The growing volume of disparate information across departments has created a need for efficient and scalable technologies to ensure data quality. Critical enterprise data are identified and linked to a relevant reference platform or central repository. This repository is the organization's trusted source (Haneem et al., 2017, 2019). MDM leads to the consistent use of more accurate and up-to-date data related to essential business entities. Although it initially emerged in the business field, this solution has also been adopted by government organizations to improve e-government services in countries such as Australia, New Zealand, the United Kingdom and Malaysia (Haneem et al., 2019). In (Vilminko-Heikkinen & Pekkola, 2017) the implementation of MDM in a Finnish municipality with 220,000 inhabitants is discussed. In (Aditya Rahman et al., 2019; Ko et al., 2021) the authors address the MDM maturity model in the Pasar Rebo public hospital within the Secretariat of the Presidential Advisory Council of Indonesia. They emphasize the importance of detecting the need to enhance MDM through the maturity model, as failure to do so can lead to the failure to achieve objectives and a loss of data consistency and quality. Each organization establishes attributes and metadata that must be considered as master data and defines how they are connected (Ericsson & Berndtsson, 2022). The domain of master data is limited in most organizations, focusing on products, clients, suppliers, etc. However, in the public sector, the focus is on citizen data and services (Haneem et al., 2017).

This work is developed within the context of MDM IT framework applied to the iDatos system, which supports statistics based on administrative records at ISTAC (Yanes et al., 2018). In particular, it is oriented to the experimentation of technological enhancements aimed at improving the performance and scalability of the system. The architecture of iDatos is built around the concept of directories. A directory groups several registers. For instance, within the population and household directory, we find the population register, and work is currently underway on the construction of the household register. The linking of analysis units across registers, either within the same directory or between different ones, generates the administrative life of each unit. The system is a living entity in which new registers are created and existing ones are updated at different intervals. For example, the PMH in the population register alone requires the update and linkage of more than 2,000,000 units in the system. The vehicle register integrates the Vehicle Fleet, Technical Vehicle Inspection, Registration, Deregistration, and Ownership Change records according to the frequency with which they are received. The housing register integrates data from the housing census, cadastral records, and other sources. It is therefore necessary to consider not only the analysis units but also their different versions and their links with others across the various registers. In short, scalability is a key requirement for the system and is expected to become a critical aspect of iDatos. It is therefore essential to have a data management system not only optimized for performance but also capable of supporting high scalability. Nowadays MDM must have the capacity to work with silos, new technologies and information sources, against later systems which had a more centralized orientation.

Data technologies are integrated into the processes of the Statistical Office. One of the challenges to address is defining the architecture, hardware, and software needed. It is essential to connect large amounts of sources and silos to obtain not only the data but also their relationships. These systems must be capable of efficiently retrieving data that require executing queries involving traversals across multiple relationships, with the associated cost in relational databases. In addition, scalability is a critical issue, as these are dynamic systems in which new administrative records are continuously integrated. In this context, modern MDM systems increasingly explore the benefits of using a Graph-Oriented Database. In contrast to these systems, Relational Databases with rigid schemas were commonly used.

A Graph-Oriented Database is a system that uses a graph data model, including methods for Create, Read, Update, and Delete (CRUD). In this model, information is represented through nodes and edges that represent relationships, which always start and end at a node. Properties can be applied to both relationships and nodes, allowing for a higher level of detail in the graph using key-value pairs. To establish categories, nodes can be labelled with one or more labels, thereby capturing the semantic roles of a data set. A Graph-Oriented Database that employs native graph processing is characterized by two fundamental properties:

- Graph Storage: This property emphasizes the representation of data as a graph of interconnected nodes and relationships, rather than as tabular structures of rows and columns. In this architecture, related nodes are sometimes even stored next to each other in memory. This proximity maximizes de probability of data is in the CPU cache, thereby reducing access latency.
- Graph Processing Engine: Direct links are established between related nodes within the database. This design enables rapid traversal of relationships, eliminating the computational overhead associated with index lookups and other retrieval mechanisms. This capability (known as index-free adjacency) allows the system to traverse millions of records per second.

In contrast, relational databases typically demonstrate lower performance in relationship traversal, primarily because they rely on index searches to locate the next related element required to complete a transaction (Robinson et al., 2015).

This type of database is particularly useful for large, highly connected datasets, where queries often require multiple joins that can degrade performance or even block the system. In these cases, performance in a Graph-Oriented Database remains constant because execution time is proportional to the length of the subgraph that responds to the query, rather than the entire graph.

Most legacy MDM systems are based on relational databases, which are not optimized for querying linked data with quick response times. Graph-Oriented Database provide an ideal solution for modeling, storing, and querying relationships between master data, metadata, and hierarchies. Additional advantages include the ability to complement existing systems without replacing them, the simplicity of the query language, the possibility of applying graph-based learning algorithms, and the ease of visualizing and communicating the data structure (Mathur, 2021).

Nowadays, there are multiple graph database projects. In (Calabuig Monerris, 2016; Das et al., 2020; Graph Everywhere, 2024) some analyses and benchmarks of the most widely used ones can be found. Four databases are highlighted in the three citepd references: ArangoDB (ArangoDB, 2024), Neo4j (Neo4j, 2024), OrientDB (OrientDB, 2024) and Titan (Titan, 2024).

These four databases support ACID transactions. Two of them (Neo4j and Titan) use native graph-based processing, while the other two (ArangoDB and OrientDB) are document-oriented databases with support for graphs

Below are the characteristics of these systems that have influenced the development of this work:

ArangoDB is a multimodal database management system because it allows the storage of documents, graphs, and data as key-value pairs. With ArangoDB, it is possible to combine different data models, and it is highly scalable, offering great flexibility. It implements its own query language, AQL (ArangoDB Query Language), and includes a web interface.

Neo4j is an open-source database management system that uses a graph-based data model. It features a distributed cluster architecture that is highly scalable. Neo4j is implemented in Java and Scala and can be used both in local installations and in the cloud with Neo4j Aura or other native cloud platforms. It also provides visual tools to facilitate querying and data management, along with RESTful APIs, Java APIs, and more. It supports the Cypher query language, which is similar to SQL but tailored for graph structures. Neo4j supports various programming languages, including .Net, Clojure, Elixir, Go, Groovy, Haskell, Java, JavaScript, Perl, PHP, Python, Ruby, and Scala. It also boasts a large community and extensive documentation.

Table 1
Positions of Neo4j, ArangoDB, OrientDB and Titan in the DB-Engines ranking.

Date	ArangoDB	Neo4j	OrientDB	Titan
October 2024	81	21	89	–
February 2025	86	20	94	–
October 2025	89	20	100	–

OrientDB is an open-source NoSQL database management system. It is multimodal, supporting documents, graphs, and objects. Implemented in Java, it allows mixed schemas and supports queries in Gremlin language as well as extended SQL.

Titan is a graph database, scalable and optimized for storing and querying graphs with millions of vertices. It is compatible with various storage backends.

The DB-Engines ranking (DB-Engines , n.d.) is a monthly classification of the popularity of database management systems. By examining the ranking positions of four graph databases at different points in time, we obtained the following results:

According to the DB-Engines ranking from February 2025 (Table 1), Neo4j is the most widely used graph data-base, ranking 20th worldwide, an improvement of one position compared to the previous year. ArangoDB and OrientDB not only rank far below Neo4j, but their positions have also declined over the past year (ArangoDB dropped by eight places, and OrientDB by eleven). Titan no longer appears in the ranking. Based on these data, Neo4j was selected for this study, while ArangoDB, OrientDB, and Titan were excluded from further consideration.

This work presents the transformation of master data management at ISTAC from a relational database schema to a Graph-Oriented Databases. A simulated test ecosystem was created to analyze the feasibility, potential issues during migration, and performance evaluation in Neo4j.

The comparison is extended to a PostgreSQL database with graph functionality, utilizing Apache AGE (A Graph Extension) (The Apache Software Foundation, 2024). Apache AGE was included in the analysis since migration to this system would likely be more straightforward for ISTAC, as it currently operates using PostgreSQL databases.

Apache AGE extends the PostgreSQL relational model by incorporating graph database functionality. While this integration enables graph-oriented operations, it does not implement native graph processing, which limits its efficiency in comparison to purpose-built graph databases. The arquitecture is built on top of a relational database rather than as a native graph engine. During database initialization, PostgreSQL generates a namespace corresponding to the graph name. Within this namespace, two relational tables are created to store nodes and relationships. Moreover, each time a new label is assigned to a node or relationship, an additional relational table is instantiated. The key advantage of Apache AGE over the traditional relational model is its aim lies in to create a unified storage system capable of managing both relational and graph models, enabling users to work seamlessly with SQL standard and Cypher queries. However, this hybrid approach entails performance trade-offs: relationship traversals are constrained by index lookups and table joins, rather than benefiting from the direct memory adjacency characteristic of native graph databases. As a result, while Apache AGE provides interoperability and flexibility, it does so at the cost speed and scalability.

Neo4j's index-free adjacency architecture relies on singly linked lists for each node, pointing to their adjacent nodes, while edges are implemented as doubly linked lists referencing both source and target nodes. Traversals are computed directly in memory using pointers, meaning that relationship lookups operate through direct memory access, and query time is proportional to the size of the explored subgraph, achieving O(1) access. In contrast, Apache AGE relies on B-tree index lookups and joins that depend on the total data size, leading to query times that grow logarithmically with the data volume, approximately O(log n).

Migrating a relational database to an object-oriented database is not a straightforward process, as it involves a significant paradigm shift. This requires rethinking the way data structures are conceptualized and migrating all the code developed up to that point.

- The main objective in a graph-oriented database is to model relationships. For this reason, relational tables are converted into nodes, while tables created to represent many-to-many relationships become edges connecting those nodes.
- Foreign keys in the relational schema are also transformed into relationships between nodes. It is essential to ensure that data integrity is preserved throughout this transformation process.
- Scripts must be developed to read the rows from relational tables and generate the corresponding nodes and relationships. For legacy databases containing large volumes of data, this process can be very time-consuming.
- Furthermore, all queries previously developed for the relational database are written in SQL. These queries will need to be rewritten using a graph query language-Cypher, in the case of Neo4j.

In addition to these technical challenges, an organization must also consider the learning curve required for its personnel to adapt to the new paradigm. Employees will need to become familiar with the new data modeling approach, the new database management system, and the new query language. Moreover, during the migration period, the relational system must remain operational. The study presented in this paper aims to support ISTAC in determining whether the performance gains achieved with the graph-oriented paradigm justify adopting this technological change within the organization.

The main contributions of this work are as follows:

- A detailed analysis of the public statistics architecture based on administrative records implemented by the Canary Islands Government.
- The design of graph-based database models for Master Data Management (MDM) aimed at integrating administrative data to support the production of official statistics.
- The implementation of four experimental databases that replicate the real environment, applying MDM with a relational, hybrid (relational + graph), and native graph database, respectively.
- The design of performance tests to evaluate each database's efficiency in the generation of statistics based on MDM.
- The analysis of different query strategies across the four experimental solutions to identify the most efficient approach.
- Recommendations for migrating a relational MDM system to one based on graph technology.

To the best of our knowledge, no previous studies have addressed the use of MDM with graph-based databases in the context of public statistics based on administrative records. This work therefore bridges a gap between three research domains-public statistics from administrative data, MDM in public administrations, and the use of graph technologies for MDM-by providing an integrated framework and experimental validation.

The rest of the document is organized as follows: Section 2 presents related works; Section 3 explains the iDatos system created by ISTAC and the original relational data model; Section 4 describes the normalized relational data model and the graph-oriented models Neo4j and Apache AGE; Section 5 outlines the results; and finally, Section 6 presents some conclusions about the work. Furthermore, Appendix A provides a selection of the results obtained from solving some problems detailed in Section 4.3, while Appendix B contains a glossary of the technical terms and abbreviations employed.

2. Related work

Different aspects related to MDM in Statistical Registers, as well as the advantages of incorporating connections between data in MDM systems-either through the use of Graph-Oriented Databases or knowledge graphs-have been explored. However, at the time of writing this work, no references to studies addressing both topics simultaneously were found.

Some works were identified that relate to the use of master data in the operations of national statistical agencies. The work (Krismawati et al., 2019) examines the maturity model of MDM (M3DM) in the Statistical Register of Enterprises in Indonesia. Although most areas are still at level 1, initial steps toward implementation have been taken. The work (Rishartati et al., 2019) deals with the analysis of the Maturity Model in geospatial data in official statistics in Indonesia, revealing that the system reaches level 5 in data protection, level 4 in data quality, level 5 in data use and ownership, and no level in maintenance.This shows an incipient stage in the MDM approach, limited to specific records, in contrast to the iDatos system, which maintains and links multiple records following the MDM model, in line with the recommendations of the UNECE Task Force. This working group has defined the relevant framework for data management in the opportunity presented to state statistical agencies as data providers rather than statistics producers. The analysis includes the role of Digital Record Information Systems, which together with technological innovation facilitate the use and reuse of data. This capacity is increased if semantic interoperability is also considered and MDM is implemented (United Nations Economic Commission for Europe, 2024).

The authors of Ericsson and Berndtsson (2022) focus on monitoring the fundamental aspects crucial for the success of an MDM program, grouped into eight categories: Vision and Business Alignment, Planning and Change Management, Governance, Information, Application, Integration and Synchronization, Migration and Implementation, and Technology. They emphasize that improving master data is not exclusively an IT project and highlight the importance of non-technical aspects while ensuring that IT-related issues are adequately addressed within the project. Similarly, in Haug et al. (2023) the authors identify four components of MDM projects: people, processes, technology, and data. The “people” dimension relates to data ownership, roles, and responsibilities. Processes focus on how and when to create, use, maintain, archive, and retire data. The “data” dimension pertains to defining master data and its relationships, while the technological dimension concerns the architecture of systems where master data is created, stored, accessed, and more. The object of study of this work is framed precisely within this technological dimension, and in its relevance to the proper functioning of MDM, through experimentation with different architectures and storage models that can provide benefits to the iDatos system.

Regarding the use of graphs in MDM, the authors of Ilonen (2023) discuss the limitations of relational databases in querying linked data with complex relationships, especially those exceeding three hierarchical levels. Relationships are not directly stored in relational databases but are derived through costly join operations. Additionally, relational data models are rigid, making it difficult to adapt to evolving data structures. In contrast, Graph-Oriented Databases offer a more flexible approach, allowing for the easy addition of new elements and relationships. The authors propose a logical graph model to validate the hypothesis that Graph-Oriented Databases are more dynamic and intuitive for master data management. They emphasize the importance of involving stakeholders in the graph design process to ensure that the model is aligned with business needs. A key finding is that the graph design should not merely replicate the relational data model but should be optimized for efficient querying and analysis. Similarly, the authors of Roy-Hubara et al. (2017) highlight the benefits of using Graph-Oriented Databases in Big Data environments with highly interconnected data. However, one issue with Graph-Oriented Databases is the difficulty of maintaining integrity constraints-for example, nodes can be connected even when their relationship does not make sense. To address this, the authors propose a two-step method for transforming relational databases into graphs. First, they adapt the ER model to a version that can be mapped to a graph, and subsequently, this intermediate model is mapped to a logical graph database schema. The adapted ER mapping resolves cases of ternary relationships, part-whole hierarchies, and inheritance, preserving original constraints, including cardinalities. These works, besides presenting a graph-based solution for MDM, share with our proposal the approach of starting from the analysis of the existing relational model and proposing improvements or transformations that result in a more effective graph design.

Some projects that use graphs in MDM rely on advanced algorithms specifically designed for graph construction, whereas the approach presented here is based on the use of an enhanced ER model for graph design. Along these lines, the authors of Ganesan et al. (2020) emphasize the importance of representing relationships in MDM using a graph. These relationships can be explicit, directly registered in the graph, but also implicit. Their work focuses on designing a Graph Neural Network algorithm to predict inherent connections. An additional example is provided in Li and Zhu (2023), which presents an enhanced fuzzy clustering algorithm for constructing graphs that improve the efficient retrieval of data from administrative records, and supports the management of statistical data across various registers.

Other proposals focus on exploiting semantics in the MDM system through knowledge graphs- in some cases to facilitate usage, and in others to infer new knowledge from the graph representation. Our current solution for iDatos is limited exclusively to graph design for the efficient exploitation of master data; therefore, the only semantics considered are those inherent in the system's nodes and relationships, without extending to a full knowledge graph or an ontology-based representation. The approach in Wang et al. (2009) highlights the need to enhance MDM by adding semantics and enabling query resolution across different domains, which requires dynamically extending concepts. The proposed system defines two ontologies: one for the core business domain and another for the external domain, using RDF and SPARQL as the query language. This solution is applied to a use case involving master data management in an insurance company. Knowledge graphs represent structured topics visually and concisely, showing natural connections between them, allowing users to find what they need more easily. The integration of data from multiple complex sources in the context of data wrangling requires solutions that simplify transforming raw complex data into more user-friendly formats. The solution proposed in Nadal et al. (2023), is aimed at less technical users, employing a graph-based framework to represent the global schema and avoid interaction through SPARQL. Key benefits include integration, encoding, and creating a query language that does not require join conditions. A crucial component is its query rewriting algorithm, which also facilitates visual representation while maintaining good performance. Knowledge graphs are a valuable resource for better leveraging data value. In (Liu et al., 2018) the authors construct a knowledge graph for the China Railway Corporation to improve the company's master data management and its applications in Big Data. After analyzing the master data types, knowledge is represented in a graph using RDF or a graph database complemented with a corresponding visualization tool. The utility of graphs for managing data from multiple Big Data sources is highlighted in Nadal et al. (2023). In this case, knowledge graphs are used to extract and merge data as a method for data governance. One of the advantages mentioned is scalability, supporting incremental design. The company's knowledge is represented in a graph used to detect business risks, allowing for the early identification of judicial, managerial, and public opinion issues. In (Vasiliev et al., 2021) the authors focus on assembling data into a semantic graph supported by RDF and SPARQL to address data integration for graph-based master data management. Specifically, they assemble data into a repository based on semantic graphs. The study compares virtualization using GraphQL against a warehouse integration layer with OntoRefine as an ETL mechanism, developing the graph in the GraphDB database. The authors find that GraphQL filtering is more limited, as semantics are confined to the server side. Conversely, RDF has limitations regarding access control without external intervention but facilitates interoperability.

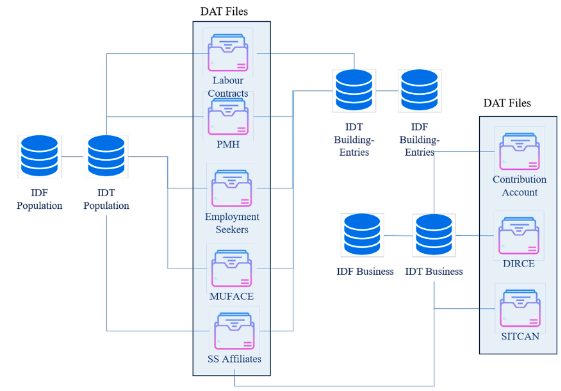

Fig. 1. iDatos architecture for the development of multi-source statistics.

3. Material

This section describes iDatos, a master data management infrastructure that supports multisource statistics and serves as the starting point for this work. The sources refer to administrative registers (e.g., PMH, social security registers, employment seekers registers) and other external sources accessed through Internet APIs to complement and enhance data quality.

iDatos is structured into four different domains:

1. Population-households.
2. Business-establishments.
3. Places.
4. Buildings-households-establishments.

These four domains are organized as follows:

- **Directories**: Conceptually related collections of re-gisters.
- **Registers**: Collections of files related to an analysis unit.
- **Layers**: Collections of register files organized by integration functionality.

Each statistical unit's variables can be categorized as:

- **Nuclear variables**: Key variables of a register.
- **Normalized variables**: Variables that are not key but must be normalized in iDatos.
- **Entity-related variables**: Identifiers used to link units in one register to those in another.

The directories included in iDatos are: Population and Households, Streets and Addresses, and Economic Units.

For simplicity, throughout the document, we refer to these directories as: Population, Building-Entries and Business, in that order. The iDatos structure is represented in Fig. 1, which illustrates how a single register can belong to multiple directories[1]

The strategy to build the registers for each directory is always the same: they are built using the reference registers (IDT) from a basic source, which is enhanced with additional auxiliary sources (IDF). The infrastructure supporting iDatos is a PostgreSQL relational database with three types of tables for each register:

[1] SS: Social Security DIRCE: Central Business Directory SITCAN: Canary Islands Territorial Information System.

Table 2
Example of 2 registers in the table IDT from directory business-entries.

uuid_idt
uuid_idt1
uuid_idt2

- DAT tables: information in administrative registers.
- IDT tables: provide a unique identifier for each record. The set of IDT tables constitutes the master database within the iDatos infrastructure, serving as the single source of truth for all data managed within a record. This data, sourced from administrative records, ensures the consistency of information for each entity involved in a statistic.
- IDF tables: complement the information in a record based on auxiliary sources. They capture each version of an entity within administrative records or big data sources involved in a statistic. The design of IDF tables includes both core variables and normalized variables that are useful for exploiting the sources to produce statistics. These files can grow as more administrative records or big data sources are incorporated into the operations.

The relations with master data are stored in URD tables for efficient exploitation in the relational database system of iDatos. These relations can be:

- **IDT to IDF**: Relations with the unique representative case and its different versions in the directory. These relations are called GEOREF in Business-Entries directories and SOURCE in all other directories.
- **IDT1 with IDT2**: Relations between elements in one directory and cases in another directory, such as people with places, people with businesses, etc. They represent the trust of one directory in another. These relations are not necessary for this work.
- **IDT to data sources or raw data** for each period or time in which they are received. These relations are called TRACE_STOCK.

In addition, additional information is collected from the relations such as the technique to link data, link type and quality, etc.

Tables 2–4 show an example of how the information is organized in the files IDT, IDF and URD from the Business-Entries directory. In the IDF tables, the unique representative case of two addresses is stored (Table 2). When the auxiliary sources are queried 14 IDF records are obtained, which are in the Table 3. Finally, to connect the 14 registers in IDF with their unique identifier (IDT) is used the URD table (Table 4). These tables have more attributes, but only the ones neccesary to understand the example are included.

This study was conducted using an anonymized sample of data from the Population, Building Entries and Business units in iDatos for June and September 2017. The sample was generated through an algorithm that guarantees the existence of linked cases. In Phase 1, a set of random records is selected in the IDT tables of each register, which we refer to as the seed. To configure it, a random point (x, y) in San Cristóbal de La Laguna is selected using simple random sampling. Then, we consider all the cases sharing the same geohash, except for the last two characters. The population living at those points, together with all the corresponding data in iDatos, constitute the initial seed. In the next step, for each case in the seed, the URD tables are queried to obtain the relations in which it is involved, either with cases in IDF tables or DAT files in the register, which are then added to the sample. These new cases, in turn, trigger the process again to obtain new relations and cases iteratively. This process allows us to establish relationships such as:

$\mathrm{IDT}_{\mathrm{DIRECTORY\_A}}$ - $\mathrm{IDF}_{\mathrm{DIRECTORY\_A}}$

Table 3
Example of 14 registers in the table IDF from directory business-entries.

uuid	Address
id1	CALLE SEÑORITA MARIA MANRIQUE LARA 2 AGAETE
id2	CALLE SRTA M MANRIQUE LARA 2 AGAETE
id3	CALLE MARIA MANRIQUE LARA 0 AGAETE
id4	CALLE MARIA MANRIQUE LARA 2 AGAETE
id5	CALLE ST M MANRIQUE LARA 2 AGAETE
id6	CALLE MANRIQUE LARA 2 AGAETE
id7	URBANIZACION RESIDENCIAL PALMITA B1 1 AGAETE
id8	URBANIZACION RESIDENCIAL PALMITA B 1 1 AGAETE
id9	URBANIZACION RESIDENCIAL PALMITA B1 2 AGAETE
id10	URBANIZACION RESIDENCIAL PALMITA B 1 2 AGAETE
id11	URBANIZACION RESIDENCIAL PALMITA B 1 3 AGAETE
id12	URBANIZACION RESIDENCIAL PALMITA B1 7 AGAETE
id13	LUGAR RS ONCE PALMITAS 3 AGAETE
id14	LUGAR RES ONCE PALMITAS 3 AGAETE

Table 4
Example of the URD table in business-entries directory.

uuid_idt	uuid_idf	rel_type
uuid_idt1	id1	GEOREF
uuid_idt1	id2	GEOREF
uuid_idt1	id3	GEOREF
uuid_idt1	id4	GEOREF
uuid_idt1	id5	GEOREF
uuid_idt1	id6	GEOREF
uuid_idt2	id7	GEOREF
uuid_idt2	idf	GEOREF
uuid_idt2	id9	GEOREF
uuid_idt2	id10	GEOREF
uuid_idt2	id11	GEOREF
uuid_idt2	id12	GEOREF
uuid_idt2	id13	GEOREF
uuid_idt2	id14	GEOREF

$\mathrm{IDT}_{\mathrm{DIRECTORY\_A}}$ - $\mathrm{Dat}_{\mathrm{DIRECTORY\_A}}$.

In a second phase, the sample was expanded to ensure it included instances of complex relationships derived from transitive links. For this study, the following relationships were included:

$\mathrm{Dat}_{\mathrm{DIRECTORY\_A}}$ - $\mathrm{IDT}_{\mathrm{DIRECTORY\_A}}$
$\mathrm{IDT}_{\mathrm{DIRECTORY\_A}}$ - $\mathrm{Dat}_{\mathrm{DIRECTORY\_B}}$
$\mathrm{Dat}_{\mathrm{DIRECTORY\_B}}$ - $\mathrm{IDT}_{\mathrm{DIRECTORY\_B}}$

By this way, probes can be done inferring relations and instances of analysis units in different registers. These data extracted from iDatos allow the recreation of a light version of the system with anonymized information for exploitation by external researchers to the ISTAC. Fig. 2 shows a diagram with Phase 1 (shaded in blue) and Phase 2 (shaded in green).

A relational database has been implemented in PostgreSQL with the same architecture as the iDatos directory, but limited to sample data. This database is made up of IDT and IDF tables from the directories: Population, Building Entries, and Business. With regard to the data sources, one table has been generated for each source, which stores data from Social Security Affiliation (DAT_Affiliation), Employment Seekers (DAT_EmpSeekers), Labour Contracts Registered (DAT_Contracts), and Social Security Contribution Accounts (DAT_Contribution) files in June and September 2017. The PMH (DAT_PMH) in July 2017 has also been included.

The relationships between the sources and master data, and between IDT and IDF, are stored in the URD table of each directory. This relational model is called ibase-source and can be consulted in Fig. 3

The distribution of data in the generated sample is presented in Table 5. It can be observed that in all directories, the volume of data in the

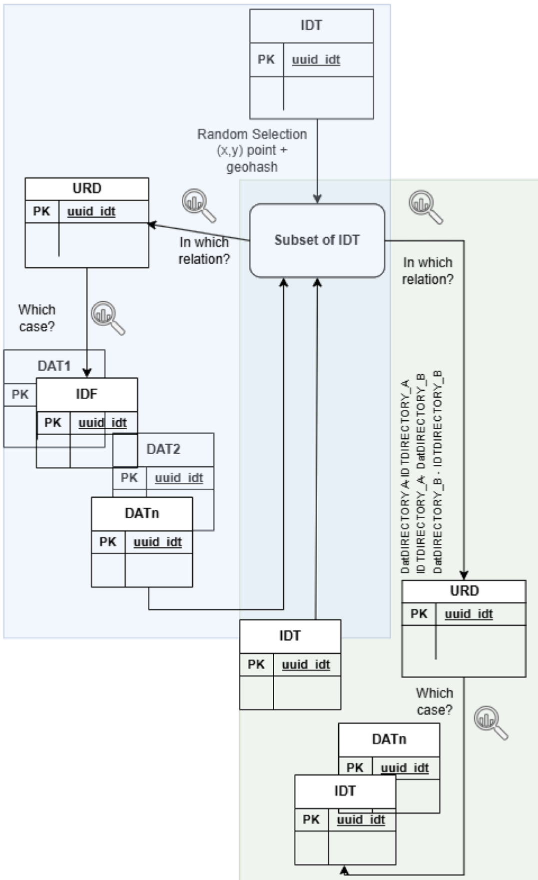

Fig. 2. Algorithm for generating the sample of cases from iDatos.

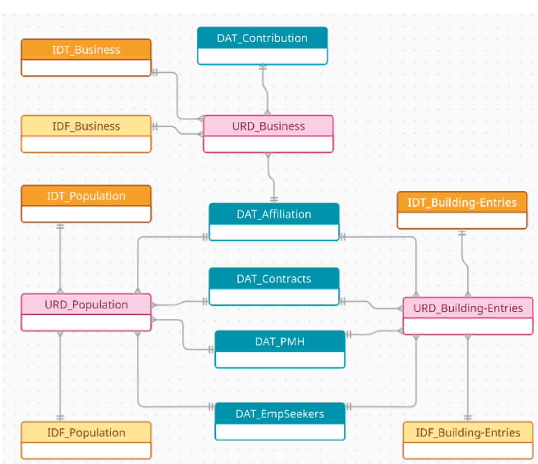

Fig. 3. Original database (*ibase-source*).The DAT elements represent more than one actual table (one table for each date).

URD tables, which represent relationships, is considerably greater than that of the other data types.

4. Methods

This section outlines the methodology employed in the research. Fig. 4 shows a conceptual research framework diagram to define the methodology followed in the development of this article. For each step, the corresponding section of the article containing additional information is indicated.

Table 5
Number of records.

Table	Number of records
IDT_Business	7815
IDF_Business	42,587
URD_Business	71,973
IDT_Population	80,265
IDF_Population	182,420
URD_Population	4,748,053
IDT_Building-Entries	73,560
IDF_Building-Entries	453,195
URD_Building-Entries	4,729,744
DAT_Contribution June 2017	89,557
DAT_Contribution September 2017	89,123
DAT_Affiliation June 2017	753,258
DAT_Affiliation September 2017	781,823
DAT_EmpSeekers June 2017	314,506
DAT_EmpSeekers September 2017	306,254
DAT_Contracts June 2017	78,843
DAT_Contracts September 2017	78,094
DAT_PMH Julio 2017	2,195,139
Total:	15,076,209

Firstly, the design of the databases used for experimentation is addressed. In addition to replicating the relational database (RDB) of iDatos, an alternative relational version was implemented, incorporating normalization improvements. Furthermore, Graph-Oriented Databases were also generated.

Secondly, the experimental procedures are presented.

4.1. Enhancement of the relational model

Certain data files within iDatos do not fully conform to the second normal form, resulting in specific attributes exhibiting multiple values. For instance, in the Employment Seekers file, attributes are defined to store language information, including language code, reading level, writing level, and other language skills. These attributes are repeated eight times to accommodate individuals with knowledge of multiple languages.

To address this issue, a second relational database (*ibase-norm*) was created. The primary modification involved partitioning the DAT_EmpSeekers table into multiple auxiliary tables. In addition, to normalizing repeated attributes, a conceptual block partitioning approach was adopted. This entailed extracting attributes and generating new tables with a one-to-one relationship to the DAT_EmpSeekers table (Fig. 5).

All newly generated tables maintain an identifying relationship with the DAT_EmpSeekers table. A numeric field was added where necessary to preserve the original order of records. A comprehensive list of the generated tables, their relationship to the DAT table, and their respective record counts is provided in Table 7.

4.2. Graph-oriented database design

iDatos representation by a graph considers nodes for representing records in IDT, IDF tables, and raw data in the files of each register. Relationships between these nodes are derived from the URD table. Labels are used to indicate the directory to which an IDT or IDF node belongs and the source file of each raw data node. The attributes of the records are mapped to node properties. A label is assigned to each relationship based on the REL_TYPE attribute in the URD tables. It should be noted that the graph is based on the normalized relational database schema. Therefore, additional nodes related to the corresponding DAT node have been generated through the AUXILIARY relationship. Each value for a given record generates an instance of the corresponding node type.

The resulting graph has defined the following types:

- Nodes: Each record in tables IDT and IDF of population, building entries and business generates an instance of the node type

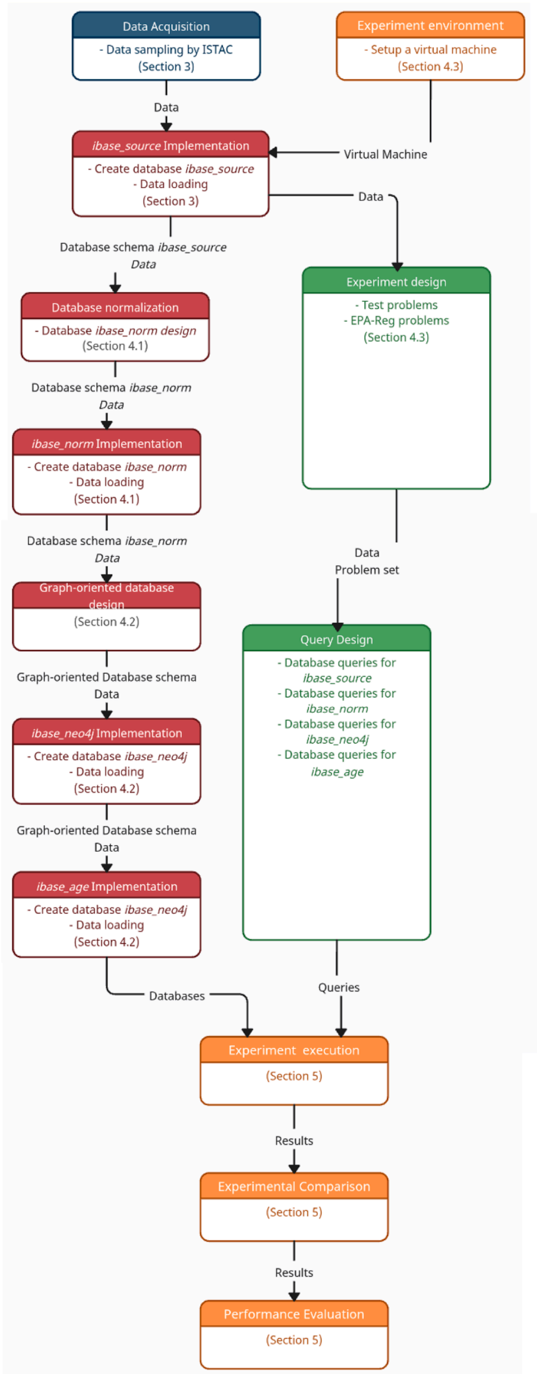

Fig. 4. Conceptual research framework diagram.

IDT_Population, IDF_Population, IDT_Building-Entries, IDF_Building-Entries, IDT_Business and IDF_Business, respectively. Similarly, several node types have been created for the registers: Municipal Register of Inhabitants, Social Security Affiliation, Employment Seekers, Contribution Accounts, and Labour Contracts registers, which are called DAT_PMH, DAT_Affiliation, DAT_EmpSeekers, DAT_Contribution, and DAT_Contracts, respectively.

- Relationships: Below are listed the cases of relationships between the node types of the graph, registered in the URD tables of each directory, along with the type of relation:
  - Directory Population and Households:
    IDF_Population → IDT_Population;

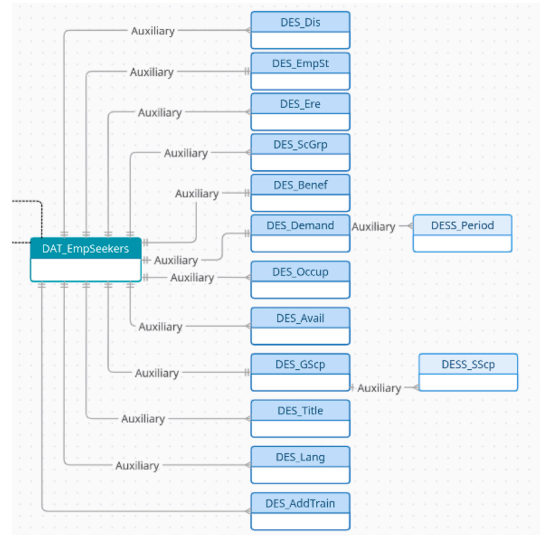

Fig. 5. Normalized relational database (*ibase-norm*).

Table 6
Number of nodes vs. number DAT records.

Type	Node number	Record number
DAT_Contribution	178,680	178,680
DAT_Affiliation	2,657,626	1,535,081
DAT_EmpSeekers	2,168,062	620,760
DAT_Contracts	374,600	156,937
DAT_PMH	3,707,517	2,195,139

    DAT_PMH → IDT_Population;
    DAT_Affiliation → IDT_Population;
    DAT_Contracts → IDT_Population;
    DAT_EmpSeekers → IDT_Population
  - Directory Streets and Addresses:
    IDF_Building-Entries→IDT_Building-Entries;
    DAT_PMH → IDT_Building-Entries;
    DAT_Affiliation → IDT_Building-Entries;
    DAT_Contracts → IDT_Building-Entries;
    DAT_EmpSeekers → IDT_Building-Entries
  - Directory Economic Units:
    IDF_Business → IDT_Business;
    DAT_Contribution → IDT_Business;
    DAT_Affiliation → IDT_Business

The schema of the graph-oriented database generated in Neo4j is shown in Fig. 6, which includes node types and relationship types.[2]

In the sample data, there are records in URD tables referencing DAT records that do not exist in the corresponding tables. As URD tables in the relational model are transformed into relationships between nodes in the graph database, it has been necessary to create the DAT nodes to establish these relationships. Therefore, we have a higher number of DAT nodes than the number of existing records in the DAT tables in the relational model (Table 6). For example, in the DAT_EmpSeeker table for June 2017, there are 314,506 records, and in the DAT_EmpSeekers table for September 2017, there are 306,254 records. This results in a total of 620,760 DAT records for DAT_EmpSeekers in Table 6.

On the other hand, in the generation of additional nodes the data number, which is necessary to maintain the existing order in the original database, has been added as an attribute of the relationship instead of

[2] DES prefix is used for Data Employment Seekers tables and DESS prefix is used for Data Employment Seekers Sublevel tables.

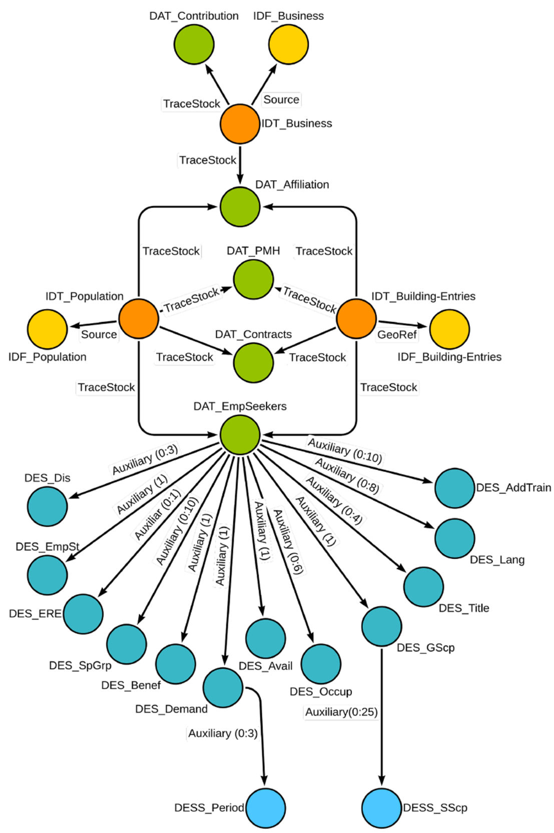

Fig. 6. Graph-oriented database using Neo4j (*ibase-neo4j*).

Table 7
Number of additional nodes vs. number of DAT records normalized *ibase*.

Content	Name	Node nº	Record nº
Disability	DES_Dis	43,869	119
Employment status	DES_EmpSt	620,760	11,584
ERE	DES_Ere	1178	1178
Special group	DES_SpGrp	271,836	10,403
Benefit	DES_Benef	620,760	620,760
Demand	DES_Demand	620,760	620,760
Demand period	DESS_Period	1,861,632	1086
Occupation	DES_Occup	2,703,084	2,703,084
Availability	DES_Avail	620,760	620,760
Geographic scope	DES_GScp	620,760	18
Search scope	DESS_SScp	805,155	805,050
Titles	DES_Title	321,232	78,097
Language	DES_Lang	362,992	1330
Additional training	DES_AddTrain	372,083	372,083

a node attribute. This design reduces the number of nodes because an additional node is related to several DAT_EmpSeekers nodes (Table 7).

Finally, to generate the graph database in PostgreSQL using the Apache AGE extension (*ibase-age*), the design explained in Fig. 6 was utilized.

4.3. Experiment design

The first step in designing the experiments was selecting the problems to solve. These were divided into two groups.

For each problem, the following information is provided: a core objetive, the specifc tables from *ibase-source* required and the number of times they are used, the output fields and a description of the operations necessary to solve it (excluding basic SQL clauses). In addition, output information for some problems is also included in Appendix A.

- Test: Designed to evaluate the performance of all databases with problems of varying complexity for comparison.

q1 Companies with documented employee reductions and corresponding reduction data between July and September 2017.
Tables:
 – URD_Business (3 times).
 – DAT_Contribution (2 times, one differentiated by date).
Output fields: uuid of IDT_Business | Reduction in the number of employees.
Operations: Parsing string field to integer value.
The number of employees in a company in a given month is calculated as the sum of employees assigned to any of the different contribution accounts associated with a company. The number of employees in June 2017 (Employees-Jun) is computed, as is the number of employees in September 2017 (Employees-Sep). The difference (Employees-Sep minus Employees-Jun) is then calculated, and companies with a positive value are excluded.

q2 Persons who were concurrently affiliates and employment seekers in June 2017.
Tables:
 – IDT_Population
 – URD_Population (2 times).
Output fields: uuid of IDT_Population.
Operations: No additional operations required.

q3 Persons with different birthdates in IDF, affiliates, employment seekers, and PMH.
Tables:
 – IDT_Population (4 times).
 – URD_Population (4 times).
 – IDF_Population.
 – DAT_Affiliation (2 times, one differentiated by date).
 – DAT_EmpSeekers (2 times, one differentiated by date).
 – DAT_PMH.
Output fields: uuid of IDT_Population | date of birth of IDF_Population | date of birth of DAT_Affiliation | date of birth of DAT_Emp-Seekers | date of birth of DAT_PMH.
Operations: No additional operations required.

q4 Persons who are in PMH but are neither affiliates nor employment seekers and are 16 years old or older.
Tables:
 – IDT_Population (4 times)-
 – URD_Population (4 times).
 – IDF_Population.
Output fields: uuid of IDT_Population | date of birth
Operations: A demographic restriction must be applied to the dataset: only individuals over the age of 16 will be included. To do this, it is necessary to calculate the age of each person based on their stored date of birth and the current system date, excluding all individuals under the age of 16.

q5 Persons with a disability who are employment seekers, have fixed-discontinuous availability, work remotely, and have at least 50 h of complementary training. The results include the number of hours of complementary training.
Tables:
 – IDT_Populations.
 – DAT_EmpSeekers (2 times, one differentiated by date).
 – URD_Population.
Output fields: uuid of IDT_Population | number of hours of complementary training.
Operations: Parsing string field into integer value.

q6 Persons affiliated with the same company who have changed addresses. The results include the number of municipalities associated with that person in affiliates, the number of addresses associated with the person, and the list of different addresses (latitude/longitude) along with the associated dates are

displayed.
Tables:
- URD_Population (2 times).
- URD_Business (2 times).
- IDT_Building-Entries.
- URD_Building-Entries.
- DAT_Affiliation (2 times, one differentiated by date).

Output fields: uuid of IDT_Business | uuid of IDT_Population | number of municipalities | number of addresses | list of month | list of latitude-longitude
Operations: To display the list of months and the list of latitude-longitude sorted by month as a single field, the *array_agg* aggregation function is used twice, applying substring function and order by clause.

- EPA-Reg: Selection of some queries established by ISTAC included in the EPA-Reg statistics. All problems in this group show historical statistical data by sex.

q7 Employed population working in a company in the same municipality of residence.
Tables:
- IDT_Population.
- DAT_Affiliation (2 times, one differentiated by date).
- DAT_Contribution (2 times, one differentiated by date).
- URD_Population.

Output fields: year | month | sex | number of people
Operations: The SQL OVER clause is used to partition the row set and separate it into two distinct subsets based on the sex field.

q8 Unemployment by duration (greater or less than one year).
Tables:
- IDT_Population (2 times).
- URD_Population (2 times).
- DAT_EmpSeekers (4 times, one differentiated by date).

Output fields: year | month | sex | number of people unemployed for more than 1 year | number of people unemployed for less than 1 year
Operations: Parsing string field into integer value.

q9 Unemployment by experience (with or without labour experience): Being a employment seeker is considered unemployed. An employment seeker is considered to have work experience if he or she is currently or has previously been an affiliate.
Tables:
- IDT_Population (2 times).
- URD_Population (3 times).
- DAT_EmpSeekers (4 times, one differentiated by date).

Output fields: year | month | sex | number of Unemployment with labour experience | number of Unemployment without labour experience
Operations: The total number of job seekers for each month, year and sex (Total Job Seekers) is computed. The calculation of the total number of job seekers is repeated, applying the restriction that a person must be in the affiliate table at the specified date, or any previous date, to be counted (Job seekers with professional expe-rience). The difference (Total job seekers minus job seekers with professional experience) co-rresponds to the number of job seekers without professional experience.

q10 Registered unemployment and employed population by economic activity sector.
Tables:
- IDT_Population (3 times).
- URD_Population (3 times).
- DAT_Affiliation (4 times, one differentiated by date).
- DAT_Contribution (4 times, one differentiated by date).
- DAT_EmpSeekers (2 times, one differentiated by date).

Output fields: year | month | cnae code (1 digit) | sex | number of registered employed | number of registered unemployment
Operations: Parsing string field into integer value.

Table 8
The average execution time in milliseconds (ms) and a multiplicative factor for the relative performance difference between PostgreSQL relational databases.

Query	Records	*ibase-source*	*ibase-norm*	factor
q1	1189	209,141.196	210,746.209	0.99238
q2	1470	2,556.723	2,528.532	1.01115
q3	382	30,179.173	27,914.667	1.08112
q4	23,150	316,021.875	310,932.623	1.01637
q5	4	7,433.221	6,998.282	1.06215
q6	9	114,771.806	112,698.768	1.01839
q7	4	9,253.361	9,025.321	1.02527
q8	4	14,651.728	16,553.782	0.88510
q9	4	38,894.181	36,841.747	1.05571
q10	40	24,015.587	22,747.818	1.05573

Several key features of the solution to these problems must be highlighted:

- The only difference between the two relational PostgreSQL databases (*ibase-source* e *ibase-norm*) resides in the DAT_EmpSeekers table, therefore query discrepancies are limited to tests q5 and q8. A comparative analysis of these discrepancies is presented in Table 8.
- Due to the inherent design of relational databases, certain PostgreSQL queries are executed exclusively on URD tables, obviating the need to access all IDT or DAT tables. However, in graph-oriented databases, it is necessary to access all nodes, a consequence of their inherent graph structure.
- When processing registers from DAT files, in relational databases query the corresponding tables and related records on the URD tables. In contrast, graph-oriented databases require the exclusion of DAT nodes generated from the relationships (Table 6).

The setup for the experimental environment utilized a virtual machine provisioned from the Infrastructure as a Service (IaaS) platform of the University of La Laguna. This virtual machine operated on the Ubuntu operating system and was configured with 32 GB of RAM, 525 GB of disk storage, and 4 virtual CPU cores. PostgreSQL, Neo4j, and the Apache AGE extension were installed on this platform.

In the implementation of SQL and Cypher queries to address the previous problems, various strategies were evaluated, including, but not límites to, the utilization of views, the WITH clause, and functions within relational databases. Consequently, for many of the proposed problem, one or more queries were formulated within the respective database environment (Tables 9, 11 and 14).

Each implemented query was executed five independent times. To ensure experimental rigor and consistent conditions, all tests were automated via a background script, with the virtual machine dedicated solely to this process.

From the five executions of each query, the minimum and maximum execution times were excluded, and the arithmetic mean of the three remaining execution times was computed. In instances where multiple strategies were executed for a single problem, the minimum recorded execution time was selected. A comprehensive overview of the results is presented in the subsequent section.

5. Results and discussion

Initially, a comparative analysis of the two PostgreSQL relational databases was conducted. Table 8 presents, for each query, the number of records returned, the average execution time in milliseconds (ms), and a multiplicative factor indicating the relative performance difference between ibase-source and ibase-norm. As observed in the results table, no significant performance disparities were identified, with the multiplicative factor ranging from 0.88510 to 1.08112.

Analysis of the two queries most affected by design differences (q5 and q8) revels that *ibase-norm* achieves superior performance for query

Table 9
Resolution strategies implemented in PostgreSQL relational databases.

Query	Strategies	Better
q2	*function / with*	*with*
q3	*function / with / view*	*function*
q4	*function / with*	*function*
q5	*function / with / view*	*function*
q7	*function / function + with*	*function + with*

q5 (factor of 1.06215), while *ibase-source* is more efficient for query q8 (factor of 0.88510). This performance discrepancy is directly attributed to the data normalization strategy. For query q5, the improved performance of *ibase-norm* is a direct result of accessing three additional tables (DES_Dis, DES_Avail, and DES_AddTrain), two of which were introduced as a result of normalization. This suggests that the normalized structure can facilitate more efficient data joins. In contrast, the superior performance of *ibase-source* for query q8 is due to its simpler and less fragmented design. Query q8 only accesses one additional table (DD_EmpSeekers), which maintains a one-to-one relationship with the DAT table (a product of conceptual partitioning). This highlights a key consideration in database design: how schema design choices directly influence query execution costs and overall system performance.

Table 9 illustrates the problems for which different resolution strategies have been implemented and identifies the most optimal strategy in each case. *Ibase-source* and *ibase-norm* are included jointly, as identical queries and results were obtained in both systems.

The different resolution strategies implemented are as follows:

- *View*: Employing a view to aggregate all DAT-type data thereby eliminating the need to access a file for each date.
- *With*: Utilizing the WITH clause to extract the necessary dataset and subsequently incorporate it into the query.
- *Function*: Implementing a function to execute a query and return the resulting record set.
- *Function + with*: Implementing a function that incorporates the WITH clause within the query to select the required data.

A comparative analysis of Tables 8 and 9 reveals that the *with* strategy outperforms the *function* strategy only for the simplest query (with an execution time of approximately 2.5 s). For *ibase-source*, in queries q3, q4, and q5, the multiplicative factor of *with* execution relative to *function* execution ranges from 1.30 to 2.05. In the final query, the multiplicative factor of function + with relative to function is 0.95, indicating a marginally better result but without significant differences. These findings are critical for query optimization. They strongly suggest that while *with* may be advantageous in trivial cases, the *function* approach is more efficient and scalable solution. This is particularly relevant for systems with complex data processing, as it provides a clear guideline for developing high-performance queries.

Table 10 presents the calculated multiplicative factors for the three PostgreSQL-based databases relative to Neo4j database. In all cases, Neo4j demonstrates superior performance, with execution times significantly lower than those of the other databases. The minimum multiplicative factor exceeds 2.4, reaching values greater than 60,000 for the relational databases and over 3000 for *ibase-age*. These results highlight Neo4j's exceptional performance for high-volume workloads, particularly when compared to traditional relational systems. This performance can translate into more efficient decision-support processes.

Table 11 presents the queries for which various resolution strategies were implemented, the minimum execution time (in milliseconds) and the identified optimal strategy.

The implemented resolution strategies are as follows:

Table 10
Multiplicative factors for the three PostgreSQL-based databases relative to Neo4j.

Query	*ibase-source*	*ibase-norm*	*ibase-age*
q1	62,748.63366	63,230.18572	1,351.50195
q2	273.94439	270.92382	2,526.67449
q3	2.96039	2.73825	7.71359
q4	22,572.99107	22,209.47307	3,434.70500
q5	21.07717	19.84388	61.74229
q6	1,446.70952	1,420.57867	202.69701
q7	3.87116	3.77576	10.49066
q8	4.00941	4.52991	6.43862
q9	2.86415	2.71301	2.42792
q10	2.62293	2.48447	131.29002

Table 11
Resolution strategies implemented and execution times in milliseconds (ms) for *ibase-neo4j*.

Query	Strategies	Time	Better
q2	*exists/matchs/with*	9.33	*matchs*
q4	*exists/optional-match*	14.00	*exists*
q5	*call/no-call*	352.67	*same time*
q7	*matchs/one-match*	2,390.33	*matchs*
q8	*call/reduce/no-call-reduce*	3,654.33	*reduce*
q9	*call/no-call*	13,579.67	*call*
q10	*call/no-call*	9,156.00	*call*

- *Exists*: Employing the EXISTS clause to verify the existence of one or more relationships.
- *Matchs*: Employing multiple MATCH clauses consecutively.
- *With*: Separating consecutive MATCH clauses by enclosing each within a WITH clause.
- *Opcional-match*: Employing an OPTIONAL MATCH clause to verify the existence or absence of a relationship between nodes.
- *One-match*: Joining multiple MATCH clauses into a single MATCH call to establish several relationships.
- *Call*: Employing the CALL clause to execute a subquery and retrieve its result.
- *Reduce*: Employing the REDUCE function to iterate over each element in a list, applying an expression, and returning the resulting value.
- *No-call / No-call-reduce*: Replacing the use of CALL or REDUCE clauses with combinations of some basic MATCH, WITH, and WHERE clauses.

Table 11 indicates that execution times exceeding 1 s are limited to four cases (q7 to q10). For queries q8 to q10, strategies utilizing Cypher-specific clauses or functions outperform those relying solely on basic clauses. The multiplicative factor of the *call* strategy relative to the *reduce* strategy is 1.63304 (q8). However, the multiplicative factor obtained when employing basic clauses ranges from 3.25008 to 8.64679. In the case of q7, both strategies yield comparable results, with a multiplicative factor of 1.01938, indicating no significant difference between establishing two relationships within a single MATCH clause or separating them into independent clauses. It is observable that calls to Cypher clauses such as CALL or REDUCE consistently demons-trate superior performance and clarity when compared to attempts to simulate the same functionality through multiple invocations of the MATCH and WITH clauses and manual result manipulation. This disparity stems from the fact that Cypher's internal functions are highly optimized for efficient data handling, leveraging index-free adjacency more directly. This capability allows a series of complex operations to be executed as a single, cohesive unit. Conversely, simulating these operations with multiple MATCH and WITH clauses often compels the query engine to repeat graph traversals or generate substantial intermediate result sets. The practical implication is that developers should exploit Cypher-specific constructs to optimize complex queries, as doing so leverages Neo4j's internal query optimizations.

Table 12
Multiplicative factors relative to the *ibase-age* database.

Query	*ibase-source*	*ibase-norm*
q1	46.42881	46.78512
q2	0.10842	0.10723
q3	0.38379	0.35499
q4	6.57203	6.46620
q5	0.34137	0.32140
q6	7.13730	7.00838
q7	0.36901	0.35992
q8	0.62271	0.70355
q9	1.17967	1.11742
q10	0.01998	0.01892

Table 13
Execution time in minutes in *ibase-source* and optimal database. A Comparison of Three PostgreSQL Database Approaches.

Query	Time	Database
q1	58.09	Graph-oriented
q2	0.71	Relational
q3	8.38	Relational
q4	87.78	Graph-oriented
q5	2.06	Relational
q6	31.88	Graph-oriented
q7	2.57	Relational
q8	4.07	Relational
q9	10.80	Graph-oriented
q10	6.67	Relational

Furthermore, the utilization of these dedicated clauses enhances query clarity. This provides a clear guideline for optimizing Cypher queries in practice.

Finally, a comparative analysis of the three PostgreSQL-based databases was conducted by calculating the multiplicative factor of the two relational databases relative to the *ibase-age* database (Table 12).

In this case, it is noteworthy that the performance varies significantly depending on the specific problem. In six of the tests, the relational databases yielded superior performance, with multiplicative factors for *ibase-source* ranging from 0.62271 to 0.01998. In the remaining four cases, the graph-oriented database provided the best results, with factors ranging from 1.17967 to 46.42881.

An analysis of query characteristics reveals that the graph-oriented database achieves better execution times for complex queries, those where execution times in the relational databases exceed half a minute. Based on the conducted experiments, the optimal database for a given problem depends on the type of operations required for its solution (Section 4.3 describes the problems and their operations). For problems that can be solved using only basic SQL clauses or simple operations (e.g. parsing strings to obtain integer values or using clauses such as OVER), a relational database offers better performance. In contrast, for problems that require extensive data manipulation, the graph-oriented database is the optimal choice.

Table 13 presents the execution times in minutes obtained with *ibase-source* for each query, along with the identification of the most efficient database type.

To work with the Apache AGE extension, queries are executed using a combination of standard ANSI SQL and openCypher (Neo4j, n.d.), employing the Listing 1.

Alternatively, Cypher queries can be embedded within WITH or WHERE clauses, and perform the main query in ANSI SQL, employing the Listing 2.

Considering these two schemas, at least two potential strategies can be employed to solve most of the proposed problems. In Table 14,

```
SELECT * FROM cypher('graph_name', $$
      Cypher Query Here
$$) AS (res1 agtype, res2 agtype);
```

Listing 1. Schema: Combination of standard ANSI SQL and openCypher.

```
 SELECT * FROM cypher('graph\_name', $$
      Cypher Query Here
 $$) AS (res1 agtype, res2 agtype)
)

SELECT * FROM name WHERE conditions
```

Listing 2. Schema: Cypher queries embedded within WITH or WHERE clauses.

the first schema is referred to as the *age* strategy, while the second schema is referred to as the *ageP* strategy. Furthermore, for certain problems, multiple resolution strategies have been applied within the same schema.

The different resolution strategies implemented align closely with those applied in *ibase-neo4j*:

- *age-matchs*: Utilizing multiple consecutive MATCH clauses.
- *age-with*: Separating consecutive MATCH clauses by placing each within a WITH clause.
- *age-one-match*: Combining multiple MATCH clauses into a single MATCH call to establish several relationships.
- *ageP-matchs*: Employing multiple MATCH clauses within an openCypher query embedded in the WITH clause of an ANSI SQL query.
- *ageP-with*: Employing multiple ANSI SQL WITH clauses to execute several openCypher queries and separate MATCH clauses.
- *ageP-exists*: Utilizing openCypher queries within the WHERE clause of an ANSI SQL query via the EXISTS operator.
- *ageP-in*: Employing openCypher queries within the WHERE clause of an ANSI SQL query via the IN operator.

For most problems, the differences between the various strategies are negligible, with multiplicative factors ranging from 1.00242 to 1.07400. Only in cases q7 and q9 do the factors exceed 1.1. Queries q7 and q9 share a similar nature, where the optimal strategy involves utilizing the *ageP* schema with multiple WITH clauses in ANSI SQL. This approach enables the execution of simpler queries on the graph, which are subsequently combined using a SQL join. The multiplicative factors range from 1.41215 to 1.76851 in q7 and from 1.89789 to 1.91588 in q9. These results highlight that the benefits of hybrid strategies (to combine the power of graph with the efficiency of relational joins) is the most effective solution for complex, multi-step problems.

Table 15 summarises the strengths, weaknesses, applicable scenarios, and the most effective strategies among those tested for different database approaches in a comparative format. The table shows that while relational approaches remain robust for simple queries, graph-based solutions such as Neo4j offer clear advantages: they enable highly efficient relationship traversals and support complex, multi-register queries with greater flexibility and scalability, making them particularly well-suited for statistics based on administrative registers. Moreover, as iDatos is an evolving system, a scalable solution is required, further reinforcing the suitability of Neo4j. However, adopting the system entails a comprehensive organizational effort, including potential challenges in staff training and adaptation to new technology.

Table 14
Resolution strategies implemented and execution times in milliseconds (ms) for *ibase-age*.

Query	Strategies	Time	Better
q1	*age / ageP*	4,504.56	*age*
q2	*age-matchs / age-with / ageP*	23,581.45	*age-matchs*
q3	*age / ageP*	78,634.93	*ageP*
q4	*ageP-exists / ageP-in*	48,085.87	*ageP-exists*
q5	*age / ageP*	21,774.47	*ageP*
q6	*age / ageP*	16,080.56	*ageP*
q7	*age-one-match / age-matchs / ageP-matchs / ageP-with*	25,076.18	*ageP-with*
q8	*age / ageP*	23,528.87	*age-p*
q9	*age / ageP-matchs/ ageP-with*	32,970.40	*ageP-with*
q10	*age / ageP*	1,202,091.47	*ageP*

Table 15
Comparison between ER, ER-AGE, and Neo4j.

	ER	ER-AGE	Neo4j
Strengths	Widely used, robust in constraint control	Interoperability	Efficient relationship traversals through index-free adjacency, scalability
Weaknesses	Relationship traversals through costly JOINs	Relationship traversals more expensive than in native graph processing	Smaller community and difficulties when introducing changes in organizations
Relationship handling	Need to cache relationships in tables	Relational tables that record the relationships	Relationships stored as direct links and close in memory
Scenario	Statistics requiring a small number of JOINs and simple operations	Statistics exploiting multiple file types across different registers that must connect with other SQL systems, complex queries	Statistics requiring linking different registers and/or multiple files within the register, all types of queries
Strategy for complex queries	Function	ANSI SQL with WITH statements + Cypher	Specific Cypher functions or clauses

6. Conclusion

The production of official statistics based on administrative registers has been established as the recommended practice in this field, with the exigency of assuring data quality. MDM has been proven to be a recommended methodology and architecture to achieve this requirement. However, this presents a technological challenge for official statistics agencies, due to the complexity of handling schemas and the scalability problems to which they are always subject. This requires a deep study of the technological aspects to establish a scalable and flexible infrastructure.

This work presents a strong case for adopting Neo4j as a robust alternative to conventional relational database systems, such as PostgreSQL, for the management and generation of master data within the iDatos infrastructure. We demonstrate Neo4j's superior capability as a DBMS, highlighting its capacity in handling large-scale, interconnected datasets. A key contribution is the development of a flexible and easily adaptable design, compatible with all existing directory versions. The work establishes a foundation for scalable implementation.

Despite the initial pilot experience being limited by data volume (approximately 15 million records) due to preserve statistical confidentiality and its experimental nature, the results point to the clear advantages of graph-based storage and index-free adjacency. These results suggest that Neo4j not only enhances performance but also increases model expressivity, thereby simplifying query formulation.

Regarding the two relational databases, although there are not significant differences between them, in general, *ibase-norm* obtains better results. However, it would be necessary to consider how to undo the partitioning done for conceptual reasons and to maintain the normalization application to avoid delays when carrying out queries with 1:1 relationships.

With respect to the database using Apache AGE, although only shorter times were obtained in 40 % of the performance tests compared to the relational databases, the results indicate that it is more efficient for queries that require higher computational time (queries that require at least half a minute of computation to be resolved in relational databases) and perform more complex operations. However, when compared to the Neo4j database, it can be seen that Apache AGE's hybrid approach leads to a reduction in performance: relationship traversals are limited by index searches and table joins, rather than benefiting from the index-free adjacency feature of native graph databases. As a result, although Apache AGE provides interoperability and flexibility, it does so at the expense of speed and scalability.

Overall, this work highlights the practical relevance of graph-oriented DBMSs in large-scale data infrastructures and provides a pathway for further optimization in both efficiency and analytical capabilities. Future efforts should focus on validating these advantages under production-level workloads, extending the approach to heterogeneous data ecosystems, and exploring domain-specific applications where graph-based models can unlock deeper insights and more advanced decision-support capabilities.

Based on the empirically obtained results, the primary recommendation is the adoption of Neo4j. However, an organization may prefer not to confront the inherent challenges associated with migrating to a completely different system (Section 1). In the case of continuing to utilize PostgreSQL, the graph-oriented database extension Apache AGE is recommended for problems that necessitate complex data operations that extend beyond the scope of basic SQL clauses or simple aggregate operations (specifically, those exhibiting execution times exceeding thirty seconds, as observed in Tables 12 and 13). Otherwise, maintaining the exclusive use of the relational database is the preferred approach. Should the ultimate decision be to retain the native PostgreSQL system, it will be essential to examine whether an increase in the number of records-particularly when handling the substantially larger data volume-significantly impacts the performance of these three PostgreSQL-based configurations.

Data availability

Data will be made available on request.

CRediT authorship contribution statement

Luz-Marina Moreno-De-Antonio: Conceptualization, Investigation, Methodology, Formal analysis, Visualization, Writing – original draft, Writing – review & editing; **Jesús-Alberto González-Yanes:** Conceptualization, Resources, Writing – review & editing; **Isabel Sánchez-Berriel:** Conceptualization, Investigation, Methodology, Formal analysis, Visualization, Writing – original draft, Writing – review & editing; **Rafael Betancor-Villalba:** Conceptualization, Resources, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Output information from problems

This appendix presents a selection of the results obtained from solving some problems detailed in Section 4.3. Table 8 presents, for each problem, the total number of records returned.

Table identifiers (*uuid* fields), which are 36-character strings, are replaced with text "uuid" and a numeric value to accommodate column width limitations.

q1. Companies with documented employee reductions and corresponding reduction data between July and September 2017.

uuid IDT_Business	Reduction
uuid1	−1
uuid2	−2
uuid3	−8
uuid4	−1
uuid5	−13
uuid6	−2
uuid7	−16
uuid8	−1
uuid9	−1984
uuid10	−1
uuid11	−2
uuid12	−1
uuid13	−65
uuid14	−1
…	…

q3. Persons with different birthdates in IDF, affiliates, employment seekers, and PMH.

uuid IDT	Birthdate IDF	Birthdate Affiliation	Birthdate Emp_Seekers	Birthdate PMH
uuid1	04OCT1956	14OCT1956	04OCT1956	14OCT1956
uuid1	14OCT1956	14OCT1956	04OCT1956	14OCT1956
uuid2	04JUN1959	06JUN1959	04JUN1959	04JUN1959
uuid3	06JUN1959	06JUN1959	04JUN1959	04JUN1959
uuid4	25APR1964	26APR1964	26APR1964	26APR1964
uuid5	23AUG1978	23AUG1978	23AUG1987	23AUG1987
uuid5	23AUG1978	23AUG1987	23AUG1987	23AUG1987
uuid5	23AUG1987	23AUG1978	23AUG1987	23AUG1987
…	…	…	…	…

q5. Persons with a disability who are employment seekers, have fixed-discontinuous availability, work remotely, and have at least 50 h of complementary training. The results include the number of hours of complementary trai-ning.

uuid IDT_Population	Number of hours
uuid1	78
uuid2	60
uuid3	294
uuid4	50

q6. Persons affiliated with the same company who have changed addresses. The results include the number of municipalities associated with that person in affiliates, the number of addresses associated with the person, and the list of different addresses (latitude/longitude) along with the associated dates are displayed.

Company	Person	Mun	Addr	Month	Lat$Lon
uuid1	uuid10	1	2	2017-06,2017-09	28.4535$-16.28478,28.455767$-16.280793
uuid2	uuid11	1	2	2017-06,2017-09	28.492847$-16.33005,28.445069$-16.298211
uuid3	uuid12	2	2	2017-06,2017-09	28.46656$-16.255705,28.446404$-16.292722
uuid4	uuid13	1	2	2017-06,2017-09	28.473714$-16.2913,28.483909$-16.314055
uuid5	uuid14	1	2	2017-06,2017-09	28.49078$-16.31739,28.485228$-16.320099
uuid6	uuid15	1	2	2017-06,2017-09	28.488664$-16.320166,28.49063$-16.37237
uuid7	uuid16	2	2	2017-06,2017-09	28.486369$-16.316079,28.463265$-16.263247
uuid8	uuid17	0	2	2017-06,2017-09	28.951637$-13.589742,28.503233$-13.871621
uuid9	uuid18	2	2	2017-06,2017-09	28.471304$-16.279824,28.470045$-16.267103

q7. Employed population working in a company in the same municipality of residence.

year	month	sex	population
2017	6	1	3978
2017	6	2	4080
2017	9	1	4116
2017	9	2	4283

q8. Unemployment by duration (greater or less than one year).

year	month	sex	greater 1 year	less 1 year
2017	6	1	2300	3324
2017	6	2	2981	3766
2017	9	1	2257	3142
2017	9	2	2988	3797

q9. Unemployment by experience (with or without labour experience): Being a employment seeker is considered unemployed. An employment seeker is considered to have work experience if he or she is currently or has previously been an affiliate.

year	month	sex	with	without
2017	6	1	657	4968
2017	6	2	813	5940
2017	9	1	1112	4288
2017	9	2	1517	5272

q10. Registered unemployment and employed population by economic activity sector.

year	month	cnae	sex	employed	unemployment
2017	6	0	1	268	794
2017	6	0	2	121	985
2017	6	1	1	464	160
2017	6	1	2	192	163
2017	9	0	1	293	746
2017	9	0	2	147	931
2017	9	1	1	461	151
2017	9	1	2	198	157
…	…	…	…	…	…

Appendix B. Glossary

This appendix presents a glossary of the technical terms and abbreviations employed in the document:

DAT tables: Information in administrative registers.

Directories: Conceptually related collections of registers.

EIL: Labour Insertion Statistics (in Spanish, *Estadística de Inserción Laboral*), a set of statistical operations based on administrative records.

EPA-Reg: Spanish Labour Force Survey based on Registers (in Spanish, *Encuesta de Población Activa con Registros*), a set of statistical operations based on administrative records.

iDatos: a master data management infrastructure, generated and maintained by ISTAC, that supports multisource statistics and serves as the starting point for this work.

IDF tables: Complement the information in a record based on auxiliary sources. The identifier field is named *uuid_idf*.

IDT tables: Provide a unique identifier for each record. The identifier field is named *uuid_idt*.

ISTAC: Canary Islands Institute of Statistics (in Spanish, *Instituto Canario de Estadística*).

Layers: Collections of register files organized by integration functionality.

MUFACE: Social Security or General State Civil Servant Mutual Society (in Spanish, *Mutualidad General de Funcionarios Civiles del Estado*).

PMH: Municipal Register of Inhabitants (in Spanish, *Padrones Municipales de Habitantes*).

Registers: Collections of files related to an analysis unit.

URD tables: Tables that store relationships in the system.

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